

Table 1: Comparison of LLM and Human Evaluation Scores in Forget Set. Metrics: Accuracy (A), Independence (I), and Clarity (C).

Claim	LLM (A/I/C)	Human (A/I/C)
Federated Deep Reinforcement Learning for Open RAN Slicing in 6G Networks		
Federated deep reinforcement learning, in which MVNOs locally train DDPG-based bandwidth allocation policies and share model updates for weighted aggregation at a non-RT RIC, produces privacy-preserving and generalizable RAN slicing policies that maintain URLLC delay and eMBB throughput SLAs under heterogeneous non-i.i.d. conditions while enabling low-latency online inference at the near-RT RIC.	5/4/4	5/4/4
Web-Based Volunteer Distributed Computing for Handling Time-Critical Urgent Workloads		
Browser-based volunteer distributed computing using WebWorkers and a centralized cross-origin task manager can viably execute embarrassingly parallel urgent workloads without client-side software installation across heterogeneous platforms.	4/5/5	4/5/5
In web-based volunteer computing, a small minority of long dwell-time sessions complete most tasks while many short sessions complete none, making performance dependent on value sessions and imposing overhead from non-value sessions.	5/5/5	4/5/5
Checkpointing partial results, batching tasks, and asynchronously prefetching new tasks increase the amount of useful work completed during volatile browsing sessions at the cost of higher server-side load.	5/5/5	5/4/5
Tuning task granularity is essential because smaller tasks increase the likelihood of value sessions but also amplify task management and communication overheads.	5/5/4	5/5/5
For data-bound workloads, XMLHttpRequest communication can outperform WebSockets, whereas compute-bound kernels are relatively insensitive to protocol choice and their browser-executed performance relative to native code depends on algorithm characteristics and browser engine optimizations.	4/5/5	4/4/5
A topological analysis of cointegrated data: a Z24 Bridge case study		
Time-delay embeddings of Z24 Bridge natural frequency time series exhibit a toroidal topology driven by cyclic temperature effects, whereas embeddings of cointegrated residuals exhibit an open-ball topology characteristic of approximately Gaussian white-noise processes.	5/5/5	4/5/5
The nonlinear dependence of the second natural frequency on temperature leads linear cointegration to leave residual topological structure, while Gaussian process-based nonlinear cointegration trained on data that include the nonlinear regime more effectively removes these effects.	5/5/5	4/5/5
Persistent homology comparisons using Wasserstein distances on Vietoris-Rips complexes of embedded series provide a quantitative topology-based metric for assessing the effectiveness of cointegration in detrending environmental and operational variability.	5/5/4	5/4/4

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Claim	LLM (A/I/C)	Human (A/I/C)
Predicting Companies' ESG Ratings from News Articles Using Multivariate Timeseries Analysis		
Annual corporate ESG ratings can be inferred from news by modeling monthly multivariate timeseries that integrate weakly supervised ESG-relevance detection, sentiment ratios for relevant and irrelevant articles, and semantic topic cluster frequencies.	4/4/4	4/4/5
Formulating ESG rating prediction as a regression task over these time-series with deep convolutional architectures yields superior predictive performance compared to classification and to index-aggregation base-lines based on document-level transformer outputs.	4/4/4	4/4/4
Weak supervision using sentence-embedding similarity to ESG definitions effectively bootstraps a DistilBERT-based classifier to generate scalable ESG-relevance signals without manual keyword selection.	5/5/5	5/5/5
A large-scale news corpus aligned to thousands of US companies and annual third-party ESG scores over multiple years is compiled and publicly released to support training and evaluation of ESG prediction models.	5/5/5	4/5/5
Bayesian optimization with partially specified queries		
Bayesian optimization with partially specified queries formalizes optimization where only a subset of input variables are specified and the remainder are randomly drawn, optimizing the conditional expectation over unspecified variables, and posterior-sampling algorithms for known and unknown input distributions—incorporating GP samples and UCB-style bonuses for empirical distribution estimates under an independence assumption—admit sublinear Bayesian cumulative regret for popular kernels without exponential dependence on dimensionality.	5/4/4	5/4/4
Federated Deep Reinforcement Learning for Open RAN Slicing in 6G Networks		
Federated deep reinforcement learning, in which MVNOs locally train DDPG-based bandwidth allocation policies and share model updates for weighted aggregation at a non-RT RIC, produces privacy-preserving and generalizable RAN slicing policies that maintain URLLC delay and eMBB throughput SLAs under heterogeneous non-i.i.d conditions while enabling low-latency online inference at the near-RT RIC.	5/4/4	5/4/5
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Deep-Learning-Assisted IoT-Based RIS for Cooperative Communications		
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Claim	LLM (A/I/C)	Human (A/I/C)
Training deep neural networks on channel state information to compute RIS element phase shifts and to classify received symbols enables IoT-coordinated, low-complexity multi-RIS cooperative communications without a direct path, with reliability governed by relay placement under Rayleigh fading and path loss.	4/4/4	4/4/5
Deep Learning Based Customer Preferences Analysis in Industry 4.0 Environment		
A deep learning approach that partitions restaurant sales time series by day-of-week and holiday type, models ordinary days with stacked LSTMs, and uses MLPs to infer total holiday demand from pre-holiday signals and allocate it across holiday days effectively captures weekly seasonality and event-driven surges in nonlinear, intermittent demand while remaining computationally lightweight for real-time deployment.	3/3/3	4/3/2
Detection of agglomerate fog based on a shallow convolutional neural network		
A shallow convolutional neural network with progressively smaller convolution kernels and pooling followed by two fully connected layers effectively detects agglomerate fog in images by capturing multi-scale global and local features while reducing overfitting and computational cost relative to deeper architectures.	5/5/5	5/5/4
Partitioning an image into sub-images and aggregating sub-classification decisions via voting improves robustness of agglomerate fog detection compared with whole-image classification by mitigating the influence of fog-free regions.	4/4/4	4/5/4
Applying dropout to the fully connected layers enhances generalization performance for agglomerate fog detection compared to training without dropout.	3/4/5	4/4/5
Training with synthetic fog generated using the atmospheric scattering model produces data sufficient to learn detectors that transfer to real-world images.	4/4/5	4/4/5

Table 2: Comparison of LLM and Human Evaluation Scores in Retain Set. Metrics: Accuracy (A), Independence (I), and Clarity (C).

Claim	LLM (A/I/C)	Human (A/I/C)
V-BOINC: The Virtualization of BOINC		
Running BOINC applications inside lightweight virtual machines that embed a BOINC client enables execution on heterogeneous volunteer hosts without per-architecture porting, while providing sandbox isolation and system-level checkpointing and recovery.	5/5/5	5/5/4

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In web-based volunteer computing, a small minority of long dwell time sessions complete most tasks while many short sessions complete none, making performance dependent on value sessions and imposing overhead from non-value sessions.	5/5/5	4/5/5
Mountable dependency disks carrying application software stacks allow BOINC applications with complex dependencies to run without modifying host environments and without inflating the base virtual machine image.	5/5/5	4/5/5
Using headless VirtualBox with bridged networking and guest control APIs satisfies the need for unique guest addressing, remote management, and VM snapshotting while maintaining user transparency on Linux and macOS.	5/5/5	5/4/4
Performance differences relative to native BOINC are primarily due to virtualization-imposed resource constraints, with negligible additional overhead introduced by the V-BOINC control layer.	5/5/5	5/5/5
Periodic virtual machine snapshots preserve job progress across host interruptions with typically low storage and time overheads for CPU-bound workloads, increasing for memory- or disk-intensive tasks.	5/5/5	5/4/5
On the application of topological data analysis: a Z24 Bridge case study		
A point-count-normalized sum of Wasserstein distances between persistence barcodes provides a sampling-invariant metric for comparing manifold topology across data partitions.	5/4/4	5/4/4
Semi-supervised Text Mining for Monitoring the News About the ESG Performance of Compa..		
A semi-supervised text mining approach that uses word-embedding-expanded ESG lexicons, valence-aware sentiment scoring, and rigorous filtering to build time-varying entity-level news indicators produces signals that, in a Dutch-language news corpus of European companies, anticipate ESG rating downgrades and enhance portfolio screening.	4/4/4	4/4/5
Distributionally Robust Bayesian Quadrature Optimization		
Maximizing the expected integrand under the worst-case distribution within a χ^2 -divergence ball around the empirical context distribution defines a distributionally robust Bayesian quadrature objective that reduces to a variance-regularized risk in a high-variance regime, admits efficient inner minimization via KKT-based bisection, and can be optimized by posterior sampling with sublinear Bayesian regret under standard kernel assumptions.	4/4/3	4/4/4
Dynamic SDN-Based Radio Access Network Slicing With Deep Reinforcement Learning for ...		
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Claim	LLM (A/I/C)	Human (A/I/C)
The global two-level resource block allocation for URLLC and eMBB with inter-gNodeB borrowing, OFDMA exclusivity, per-user resource limits, eMBB minimum rates, and URLLC delay bounds constitutes a non-linear binary optimization problem that is NP-hard via reduction from the 0–1 knapsack problem.	5/5/5	5/5/4
A two time-scale SDN-based RAN slicing mechanism that allocates resource blocks to gNodeBs using a single-agent EXP3 bandit and performs per-user scheduling and cooperative inter-gNodeB resource sharing using multi-agent deep Q-learning dynamically adapts to traffic and channel variations while limiting controller–gNodeB signaling.	5/5/5	5/4/5
Packet delays for URLLC and eMBB traffic are modeled with an M/M/1 queuing system, and minimum-rate and maximum-delay constraints are enforced within the allocation and learning processes.	5/5/5	4/5/5
Support for multiple 5G NR OFDM numerologies is achieved by selecting numerology-specific trained models at the gNodeB based on channel state information.	4/5/4	4/5/5
Deep-Learning-aided Detection for Reconfigurable Intelligent Surfaces		
A deep learning detector with fully connected layers and tanh activations can jointly infer channel fading and RIS phase shifts directly from received signals without pilot signaling, enabling symbol detection in passive RIS-assisted links with lower inference complexity than maximum-likelihood and MMSE detectors and robust error performance under imperfect CSI and mismatches in channel models or RIS element counts.	4/4/4	4/4/5
Daily Sales Forecasting for Grapes by Support Vector Machine		
Support vector regression with radial basis function kernels, trained with cross-validated hyperparameters on limited historical sales augmented by quantified weather, weekday/weekend, and price information, yields more accurate daily demand forecasts for table grapes than artificial neural networks and decision trees in small-sample, nonlinear retail settings.	5/4/4	5/5/4
Fast Fog Detection for De-Fogging of Road Driving Images		
Temporal averaging of the value-to-saturation ratio computed within a vanishing-point-centered region of interest provides a fast, training-free indicator of fog presence and density in road driving images, and using this indicator to modulate transmission while detecting sky via a blue-channel threshold and processing sky (gamma) and non-sky (de-fogging) separately reduces over-enhancement and sky color distortion.	4/3/3	3/4/3